

SOA Survey

Post Training Quantization for Diffusion Models

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Abstract

Diffusion models generate images via samplers in an iterative denoising process until we generate a clean image that matches images from a real distribution or dataset. Architectures like DDPM use U-Net backbones during sampling as noise predictors to use to subtract noise from a noisy image in the reverse denoising process. However, these evaluations are usually across hundreds of timesteps and therefore are computationally heavy in terms of compute and inference time. To facilitate deployment in edge or resource-constrained environments, model quantization serves as a pivotal method for both compression and acceleration. We reviewed 8–10 papers for efficient quantization of diffusion models, specifically under a post-training setting for diffusion architectures with U-Net as a backbone. We found that two challenges still persist in the field: 1) activation distributions vary across timesteps and 2) quantization error accumulates over timesteps which leads to degradation in generated image quality. We propose a frequency-spectral based grouping approach of timesteps, along with a timestep and activation aware continuous scale function, with some treatment towards the mismatches of dynamic ranges in the U-Net for skip connections via normalization.

1. Problem Definition

Diffusion models are some of the state of the art generative models for images. They perform image synthesis through an iterative denoising process that takes a pure noisy image and we iteratively reduce noise until we generate a clean image. These models use a U-net backbone or small convolutional neural networks that try to predict how much noise was added to the original image to iteratively remove it from the noisy state. These architectures largely outperform prior networks like variational auto-encoders and generative adversarial networks (GANs) that generally suffer from producing blurry images due to how their loss functions are defined or often suffer from training instabilities that restricts the model learning process

of the representations in a given image.

In diffusion systems like denoising diffusion probabilistic models require numerous U-net evaluations for a large number of time steps usually couple hundreds and at times thousands to predict noise at a particular time-step in the denoising process. These evaluations are quite expensive as the U-net processes a 512x512 input whose internal architecture is also deep in nature. This massively increases the required compute. Moreover, another natural consequence of this is in the setting of edge deployments, where we need fast inference times for data processing and outputs to solve specific downstream tasks like image classification and generation. Requiring large number of computations by DDPM makes it not edge friendly due to high latency.

Post training quantization (PTQ) of activations to INT8 is an attractive solution to reduce inference costs, but diffusion sampling is a challenging regime because quantization error is re-injected and compounded over many denoiser calls. Moreover, activation statistics vary strongly across the denoising trajectory. This is because when we start with a pure noisy image during sampling, the representation or structure changes during the denoising path because we iteratively reduce noise and try to approach a representation similar to a clean image ideally in generation. Due to this variation, when processed by the U-net architecture, we collect statistics that are too different across time-steps. So a single static activation scale can be too small in some regions of the denoising trajectory which can produce errors and distortion in generation due to clipping when applying the quantization operator.

Our report performs a thorough review of literature, specifically targeting diffusion architectures with U-

net backbones, identifying these problems, the current limitations across different proposed methods, and how we plan to solve them.

2. Literature Review

Post-training quantization (PTQ) has emerged as a practical compression strategy for deploying large-scale models without the cost of retraining. In the context of diffusion models, however, PTQ presents unique set of challenges that go beyond standard quantization settings:

2.1. Challenge 1: Varying activation distributions across time steps

Diffusion models generate images through an iterative denoising process, where at each timestep t , the model approximates the posterior distribution $p_\theta(x_{t-1}|x_t)$. Concretely, a noisy image x_t and the current timestep t are passed to a U-Net, which predicts the noise $\epsilon_\theta(x_t, t)$; the next latent x_{t-1} is then obtained via the reparameterization trick:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z, \quad z \sim \mathcal{N}(0, I) \quad (1)$$

Because the input x_t transitions from pure Gaussian noise at $t = T$ toward a structured, clean image at $t = 0$, the internal representations processed by the U-Net shift dramatically across the denoising trajectory. This produces activation distributions that are highly non-stationary — statistics collected early in sampling look fundamentally different from those collected late.

Because the input x_t transitions from an isotropic Gaussian noise at $t = T$ toward a structured, clean image at $t = 0$, the internal representations processed by the U-Net shift significantly across the denoising trajectory. This produces activation distributions that are highly non-stationary. Statistics collected early in sampling look fundamentally different from those collected late. For example, early in the sampling process, we would observe much high frequency features because most of the image consists of noise. However, as we move towards cleaner distributions, the

frequency of features now change. We would observe mixtures of different frequencies representing things like texture, edges, smoothness, blur, color variation, object shapes, background patterns, and so on.

This poses a direct problem for post-training quantization. Standard PTQ methods compute a single static activation scale from calibration data and apply it uniformly at inference. When activation distributions vary as across diffusion timesteps, a fixed scale that is appropriate for one region of the trajectory will be too small for another, causing clipping of large activations and introducing quantization error that accumulates across steps. The result is visible degradation in generation quality.

Q-Diffusion highlights this specific issue and proposes time-step aware calibration methods. Instead of taking a small calibration set and computing the required statistics for quantization and computing the scale, they run all forward passes across the full denoising trajectory and collect activations at different timesteps so that quantization scales account for the full distribution of activations the U-Net sees not just a random small set from the full data distribution. However, the error accumulation issue is not explicitly modeled and they struggle in pushing quantization for both weights and activations. They have strong results for quantization of activations or weights only, but not both. Another limitation is that the time-step aware scale idea is a little misleading. This is because in the end they still produce a static scale and one static scale to account for all the activations for all time-steps is difficult.

PTQ4DM’s proposal was Normally Distributed Timestep Calibration (NDTC). It meant sampling calibration timesteps from a skewed normal distribution rather than uniformly. The reasoning behind this is that the denoising trajectory is not uniform. Early timesteps and late timesteps produce the most extreme and distinct activation distributions, so a skewed distribution oversamples these regions of the trajectory where a single static scale is most likely to fail. They achieved 8-bit quantization. However, their timestep sampling remained was arbitrary and experiments were limited to small datasets at low resolution, leaving 4-bit quantization and high-resolution text-to-image generation unsolved.

Alternatively, APQ-DM targets two particular limitations in PTQ4DM and Q-Diffusion: heuristic timestep sampling and a single shared quantization function across all timesteps. APQ-DM implements a distribution-aware quantization function for activation discretization in different timesteps, partitioning timesteps into groups with importance weights optimized by differentiable search algorithms, and selects the optimal timesteps for informative calibration image generation using the structural risk minimization principle (SRM). The idea of SRM is that rather than picking calibration timesteps by hand or uniformly, it selects timesteps that minimize the upper bound on true generalization error. Dividing the timesteps into groups reduces clipping and rounding errors for varying activations. A limitation that the authors of this paper acknowledge is that it still produces static scales within each group. So, the issue is still not fully resolved.

Temporal Dynamic Quantization’s insight is that rather than computing a single static scale offline, the quantization interval itself should be a function of time. They propose a learnable small network called the TDQ module that predicts the quantization interval by taking the current timestep as input, uses a Fourier encoding of the timestep to evaluate activations collectively across multiple timesteps, making the learning process more stable. The reasoning is that a small MLP conditioned on t can implicitly learn the mapping from timestep to appropriate scale. Moreover, their approach has no computational overhead during inference because the timestep-dependent intervals are pre-computed and stored as lookup tables before deployment or inference. TDQ explicitly flags that input-dependent dynamic quantization (gathering statistics on a given activation at runtime) introduces notable overhead, and may not be practical. TDQ utilizes an MLP layer to estimate the activation quantization parameters for each step meaning it is not truly training-free and adds learnable parameters on top of the base model. They also do not address weight quantization. The method targets activations only, leaving W4A4 quantization not that explored.

SmoothQuant is not a diffusion paper, but it directly addresses the core difficulty that all diffusion PTQ papers face: outliers in activations dominate the maximum magnitude measurement, leading to very low

effective quantization bits or scales for non-outlier channels. SmoothQuant resolves this by transferring the quantization difficulty from activations to weights through a mathematically equivalent transformation specifically, they divide each activation channel by a per-channel smoothing factor S and absorb S into the preceding weight matrix. This makes both weights and activations numerically well-behaved and jointly quantizable to INT8. The limitation for diffusion models is that SmoothQuant computes smoothing factors from a static calibration set. So it assumes that outlier patterns are consistent across the activations. In diffusion U-Nets, the outliers itself shifts across timesteps t meaning a single set of smoothing factors computed offline cannot account for the full trajectory. However, if we implement this in a time-aware manner we may be able to see some gains when we later discuss our method.

2.2. Challenge 2: Quantization Error Accumulates Across Timesteps

A less obvious but another challenging problem in diffusion model quantization is that errors do not stay isolated at the timestep where they occur. they carry forward through the entire denoising trajectory. Because the output x_{t-1} of one denoising step becomes the input to the next, a quantized model that introduces even a small per-step error is accumulated over hundreds of steps.

PTQ4DM and Q-Diffusion both implicitly assumed that reducing per-step quantization error through better calibration was sufficient to cater this issue. Q-Diffusion’s timestep-aware calibration does reduce per-step error as a side effect, meaning less error accumulates at each step but does not model the dependency between steps. Their scales are computed assuming clean full-precision inputs at every timestep, but once quantization pushes x_t slightly off the true trajectory, subsequent inputs no longer match the distribution seen during calibration. Q-Diffusion acknowledges this explicitly as a limitation.

PTQD was one of the early papers that explicitly addressed this issue. Rather than trying to minimize per-step error at calibration time, they model quantization noise analytically at each denoising step. They decompose the noise into a correlated component

which can be estimated and subtracted and a residual uncorrelated component. The reasoning is that if you understand the statistical structure of quantization noise, you can correct for it directly rather than hoping good calibration would solve the problem. The limitation is that their correction relies on estimated statistics from a small calibration set, so if the calibration distribution is not representative or not diverse enough, the correction is not the best.

TAC-Diffusion identifies two causes for the accumulation: error propagation and exposure bias. Error propagation is the compounding feedback loop described above. Exposure bias is subtler — during calibration the model always sees clean full-precision inputs, but at inference it sees its own quantized corrupted outputs, meaning it was never calibrated for the distribution it actually receives at runtime. TAC-Diffusion addresses both through two components applied dynamically at every step: Noise Estimation Reconstruction (NER), which corrects the corrupted noise estimate ϵ_θ directly, and Input Bias Correction (IBC), which estimates the average drift the quantized model introduces per timestep during calibration and subtracts it at inference. The limitation is that IBC corrects for the mean drift only. It does not account for sample-specific deviations, so as quantization becomes more aggressive, the correction becomes less reliable.

3. Research Gap and Proposed Direction

The reviewed literature shows that activation distributions in diffusion models vary substantially across timesteps, and while standard PTQ methods have proposed meaningful solutions, limitations persist in each approach that result in measurable degradation in image generation quality. Several works have attempted to address this through timestep-aware calibration and grouping strategies. However, four concrete gaps remain unaddressed.

First, existing grouping strategies are purely empirical: they partition timesteps by index or by clustering activation statistics, without relating the grouping criterion to any structural property of the denoising process itself. The denoising trajectory has a spectral structure where early timesteps consist of low-frequency content while later timesteps progressively

recover high-frequency detail, yet no existing PTQ work exploits this structure as a basis for partitioning timesteps into quantization groups.

Second, all existing timestep-aware quantization methods (including PTQ4DM, Q-Diffusion, APQ-DM, and even TDQ) ultimately produce a static or piecewise-static scale. PTQ4DM, Q-Diffusion, and APQ-DM compute fixed scales from calibration data, with timestep information influencing only which data is collected in calibration. TDQ improves on this by learning $s = f(t)$, a continuous function of the scalar timestep index, but conditions solely on t rather than on the actual activation distribution at that step. This is an issue because the ultimate objective of quantization is to ensure that the dequantized activations remain as close as possible to their full-precision counterparts. Conditioning the predicted scale jointly on the timestep and the sample-specific activation statistics, with a loss that minimizes the discrepancy between FP32 and dequantized activations, would therefore be more principled. No existing method produces a scale jointly conditioned on the timestep and sample-specific activation statistics, leaving the static scale limitation only partially addressed in the most recent literature.

Third, the smooth-quantization migration hyperparameter α which controls how much of the quantization difficulty is shifted from activations to weights has never been applied to the diffusion PTQ setting. Existing works relating to SmoothQuant treat α as a fixed constant, allocating the activation-weight burden split uniformly across all timesteps and layers. This is particularly problematic in diffusion U-Nets where the relative difficulty of quantizing activations versus weights changes across the denoising trajectory: a fixed α appropriate for one spectral regime may be suboptimal for another. No existing work optimizes α per timestep group, making this an open and explorable direction.

Fourth, the mismatched dynamic ranges of skip connection paths and main paths within the U-Net introduce a structural source of quantization error. Q-Diffusion identifies this error and addresses it through split quantization, quantizing each branch separately before concatenation. However, this approach does not eliminate the underlying range mismatch. Techni-

cally, two tensors with separately allocated scales are being concatenated, which introduces inconsistency in the merged representation. No existing work proposes to normalize skip and decoder paths to a shared dynamic range prior to the concatenation point, which would allow a single unified scale to be applied to the concatenated tensor at near-zero additional inference cost.

Of these gaps, we propose computing the 2D spatial FFT of the noisy signal x_t at each timestep and grouping timesteps by the dominant frequency content of their power spectral density, hypothesizing that the spectral decomposition of x_t reflects the underlying representational regime of the denoising process more faithfully than activation statistics or timestep indices alone. Building on this grouping, we directly address the static scale limitation by training a lightweight MLP per spectral group that produces a continuous, sample-aware scale function $s(t)$, taking as input the scalar timestep t , per-channel activation statistics $(\mu, \sigma, \max)$, and the collapsed PSD feature vector, and outputting a per-channel scale vector optimized to minimize quantization reconstruction error within that group. This replaces piecewise-static scales with a smooth function that can respond to both the timestep and the sample-specific activation distribution, directly addressing the gap left open by TDQ. The MLP-predicted continuous scale is then plugged into a SmoothQuant-style migration, addressing the third gap by making the migration hyperparameter α learnable per spectral group: rather than a fixed global rule, we optimize α_g at calibration time by minimizing dequantization error within each group, allowing the activation-weight burden split to adapt to the distinct quantization characteristics of each frequency regime. For the fourth gap, we propose normalizing skip connection and main path tensors to a shared dynamic range before the merge point by absorbing per-path scale factors into upstream weights offline, a zero-runtime-cost re-parameterization that structurally reduces the quantization error introduced by mismatched tensor ranges at skip connections. Together, these contributions form a unified PTQ framework that addresses the principal failure modes identified in the reviewed literature.

Prior work identifies timestep-dependent activation variability as the central challenge of diffusion PTQ

but addresses it with grouping heuristics and fixed or scalar-conditioned scaling rules that lack grounding in the structure of the denoising process; we therefore propose to investigate whether grounding timestep grouping in the spectral frequency content of x_t , jointly with per-group continuous scale learning, per-group migration parameter optimization, and offline skip-connection normalization, yields a more principled and effective PTQ framework for diffusion models.

Research Question: Can the progressive shift in spectral frequency content across the reverse diffusion trajectory serve as a principled, calibration criterion for partitioning timesteps into quantization groups, and does combining spectral grouping with per-group continuous scale learning via a lightweight MLP, per-group smooth-quantization migration, and offline skip-connection normalization improve quantization fidelity over empirical grouping baselines under a post-training quantization setting?

4. Benchmark, Evaluation, and Data Landscape

Table 1: Key datasets in the field.

Dataset	Size	Modality	Notes
CIFAR-10	60k	RGB Images	Standard baseline for 32×32 resolution.
CelebA-HQ	30k	RGB Images	High-quality facial benchmarks (256×256).
LSUN Bedroom	3M	RGB Images	Testing structural and architectural texture.
MS-COCO	330k	Text-to-Image	Evaluates prompt alignment in Latent Diffusion.
ImageNet	1.2M	RGB Images	Evaluation for class-conditional generation.

Evaluation Protocol. The most widely accepted metric to evaluate the performance of a generative network is the Frechet Inception Distance (FID), which utilizes an Inception V3 model. It compares the pixel-space distribution of the model to the desired test set. Often, as a secondary reference, an older, less stable metric is used, in the form of the Inception Score (IS); though it is primarily utilized for the ImageNet dataset.

To measure the computational efficiency of a model, the Bit Operations (BOPs) are calculated for a single forward pass of the diffusion model. For a given layer requiring M multiply-accumulate (MAC) operations, and the bit-width of both the weights (b_w) and the

activations (b_a) the BOPs count is formulated as $M \times b_w \times b_a$.

Compute Context. In the surveyed literature, calibration and inference latency experiments are predominantly conducted on single NVIDIA RTX 3090 (24GB) or RTX 4090 (24GB) GPUs to demonstrate feasibility on high-end consumer hardware.

Experimental Design Considerations

4.1. Baselines

To demonstrate a meaningful contribution, the proposed framework must be evaluated against the following specific baselines to isolate improvements across each core component of the methodology:

U-Net models It looks at the whole denoising timeline to set scales (timestep-aware calibration) and uses a basic split method for skip connections. Beating papers like Q-Diffusion proves that the proposed framework improves upon the basic assumption that full-trajectory calibration is sufficient.

APQ-DM (Towards Accurate PTQ) This method uses the structural risk minimization (SRM) principle to select optimal calibration timesteps. Outperforming APQ-DM is required to validate the hypothesis that spectral frequency content is a better grouping criterion compared to activation statistics.

Temporal Dynamic Quantization (TDQ) This paper uses a lightweight MLP as a scale function; however, it is only conditioned on the timestep. We must beat this baseline in order to prove that conditioning the MLP on sample-specific activation statistics ($\mu, \sigma, \max$) significantly reduces dequantization error over relying on the timestep index alone.

SmoothQuant This method makes quantization easier by shifting the mathematical burden from the activations to the weights using a fixed number (α). We will compare against it to show that learning a flexible, unique α_g for each of our frequency groups works much better.

4.2. Meaningful Ablation Studies

An ablation study must disentangle the proposed pipeline to isolate the impact of the four primary contributions. The following variables should be isolated:

Grouping Criterion (Spectral vs. Statistical) We will swap out our custom frequency grouping (2D FFT) for a basic uniform grouping or APQ-DM’s SRM grouping. This proves if grouping by frequency performs better than grouping via activation statistics.

Scale Function Inputs Within the MLP, ablate the per-channel activation statistics ($\mu, \sigma, \max$) so the network only receives the scalar timestep t and the collapsed PSD feature vector. This will show exactly how much value comes from being “sample-aware” compared to TDQ’s older method.

Testing our Flexible SmoothQuant Forcing our α_g parameter to be a single fixed number across all groups, instead of letting it adapt, isolates whether having a flexible α_g for different frequencies actually helps.

Testing our Skip-Connection Fix Instead of the proposed offline normalization of skip and main paths, we return to Q-Diffusion’s older method of splitting them. This tests whether a single unified scale on a shared dynamic range structurally reduces concatenation error compared to applying separate scales.

4.3. Evaluation Pitfalls

The papers reviewed warn of a few common mistakes to avoid when testing:

Ignoring Accumulated Errors Evaluating only the per-step reconstruction error is highly misleading due to exposure bias; the model sees clean inputs during calibration but its own quantized, corrupted outputs at inference. The final, finished image quality must be measured using standard metrics like the Fréchet Inception Distance (FID).

Forgetting About Speed Because we are adding an MLP to calculate scales at runtime, we might accidentally make the model too slow. We must measure

the actual math operations (BOPs) and run times on GPUs like the RTX 3090 to prove our method is still fast enough for real-world use.

Using Bad Calibration Data: If the images we use to calibrate the model don't have enough variety, our error corrections will fail on new images. We must test on a diverse set of datasets, like LSUN for textures, ImageNet for classes, and MS-COCO for text prompts.

5. Annotated Reading List

Ho, J., Jain, A., and Abbeel, P. "Denoising Diffusion Probabilistic Models." NeurIPS 2020.

Proposes diffusion models as a new paradigm for generative modelling, introducing the forward noising process and the reverse denoising process learned by a U-Net.

It is the foundational architecture we want to explore under a quantization setting, as all diffusion PTQ methods including ours build directly on top of the DDPM framework.

No specific limitations with respect to quantization are discussed, as quantization of diffusion models was not a concern at the time of publication.

Wang, Y., et al. "Diffusion Model Quantization: A Review." 2024.

Conducts a thorough in-depth literature review of quantization of diffusion models, covering limitations, challenges, recent works, and quantization of different diffusion architectures including U-Net and DiT-based models.

Helps identify open challenges in the field, provides direction on what has been attempted, and offers useful context on experimental settings and evaluation protocols adopted across the literature.

As a review paper, limitations are difficult to assess directly; however, coverage of newer directions such as spectral or frequency-aware quantization strategies is absent.

Li, X., Liu, Y., Lian, L., et al. "Q-Diffusion: Quantizing Diffusion Models." ICCV 2023.

Proposes a post-training quantization method for diffusion models, introducing timestep-aware calibration across the full denoising trajectory and split shortcut

quantization to handle bimodal activation distributions in U-Net skip connections.

Directly relevant as the foundational U-Net PTQ paper this work builds upon; both the timestep-aware calibration strategy and the skip connection treatment are precursors to our proposed contributions.

Not truly timestep-aware at inference since it still produces one global static scale; the split shortcut fix is hand-engineered for U-Net architecture and does not address the underlying dynamic range mismatch.

He, Y., et al. "PTQD: Accurate Post-Training Quantization for Diffusion Models." NeurIPS 2023.

Models quantization noise analytically at each denoising step, decomposing it into a correlated component that can be estimated and subtracted and a residual uncorrelated component that is absorbed by recalibrating the diffusion variance schedule.

Relevant as the first work to explicitly treat error accumulation across timesteps as a first-class problem rather than an implicit side effect of calibration, directly informing our understanding of Challenge 2.

The noise correction relies on statistics estimated from a small calibration set, so if the calibration distribution is not representative the correction degrades.

Shang, Y., et al. "Post-Training Quantization on Diffusion Models." CVPR 2023.

Proposes Normally Distributed Timestep Calibration (NDTC), sampling calibration timesteps from a skewed normal distribution to oversample early and late timesteps where activation distributions are most extreme and a single static scale is most likely to fail. Shows how better calibration that is timestep-aware can reduce quantization error, reduce trajectory drift, and partially address the varying activation distribution problem which is a direct build up from Q-diffusion. Oversampling early timesteps produces scales biased toward early regions of the trajectory, which may not generalize well to later timesteps and can lead to rounding errors or clipping experiments are also limited to small datasets.

Xiao, G., Lin, J., Seznec, M., et al. "SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models." ICML 2023.

Identifies that activation outliers make joint weight and activation quantization difficult, and proposes a mathematically equivalent per-channel smoothing transformation that migrates quantization difficulty from activations to weights by dividing activations by a smoothing factor s and absorbing s into the preceding weight matrix.

Directly relevant as the foundation of our per-group SmoothQuant migration idea; we extend their fixed global α to a learnable per group α_g to account for the timestep-varying activation-weight difficulty split in diffusion U-Nets.

The smoothing factor s is computed once from a static calibration set and is fixed at inference, which breaks down in diffusion U-Nets where the outlier structure itself shifts across timesteps t and a single static s cannot account for the full trajectory.

Wang, X., et al. "Towards Accurate Post-Training Quantization for Diffusion Models." CVPR 2024.

Proposes timestep grouping with group-wise quantization functions for activations, selecting optimal calibration timesteps using the structural risk minimization principle to minimize the upper bound on generalization error rather than relying on heuristic uniform or skewed sampling.

Directly relevant to our spectral grouping idea as the most principled existing grouping baseline. Our work challenges their grouping criterion by proposing a criterion based in the spectral frequency content of x_t rather than activation statistics.

Grouping is still based on activation statistics with no basis in the representational structure of the denoising process. Each group still produces a static scale, and the differentiable group search adds calibration overhead.

So, J., et al. "Temporal Dynamic Quantization for Diffusion Models." NeurIPS 2023.

Proposes a learnable MLP that takes a geometric Fourier encoding of the scalar timestep t as input and outputs a continuous quantization interval, pre-computed into a lookup table at calibration time so that no computational overhead is introduced at inference.

Directly relevant to our per-group MLP scale idea. TDQ demonstrates that a continuous scale function

over the trajectory is feasible, but our work extends this by conditioning the scale on both t and sample-specific activation statistics rather than on t alone.

The scale is conditioned only on the scalar timestep index $s = f(t)$ and not on the actual activation distribution at that step, meaning two different samples at the same timestep receive identical scales regardless of their activation magnitudes, leaving the static scale limitation only partially resolved.

References

- [1] J. Ho, A. Jain, and P. Abbeel. Denoising Diffusion Probabilistic Models. *NeurIPS*, 2020. *What it does: proposes diffusion models as a new paradigm for generative modelling, introducing a forward noising process and a reverse denoising process learned by a U-Net. What it contributes: establishes the DDPM framework and the U-Net noise estimation architecture that all early subsequent diffusion PTQ methods build upon. Why it is here: it is the foundational architecture this work targets under a post-training quantization setting.*
- [2] Y. Wang et al. Diffusion Model Quantization: A Review. *arXiv preprint*, 2024. *What it does: conducts a thorough literature review of quantization of diffusion models, covering challenges, recent works, and quantization of different architectures including U-Net and DiT-based models. What it contributes: highlights open challenges and experimental settings across the diffusion PTQ literature. Why it is here: help identify open problems and research directions that directly motivated the gaps addressed in this work.*
- [3] X. Li, Y. Liu, L. Lian, H. Yang, Z. Dong, D. Kang, S. Zhang, and K. Keutzer. Q-Diffusion: Quantizing Diffusion Models. *ICCV*, 2023. *What it does: proposes timestep-aware calibration across the full denoising trajectory and split shortcut quantization to handle bimodal activation distributions in U-Net skip connections. What it contributes: first method to achieve 4-bit weight quantization of diffusion models with FID degradation of at most 2.34 compared to over 100 for naive PTQ. Why it is here: the foundational U-Net PTQ paper this work builds*

upon both its timestep-aware calibration strategy and skip connection treatment are direct prior related works to our proposed contributions.

- [4] Y. He et al. PTQD: Accurate Post-Training Quantization for Diffusion Models. *NeurIPS*, 2023. *What it does: models quantization noise analytically at each denoising step, decomposing it into a correlated component that is subtracted and a residual component absorbed by recalibrating the variance schedule. What it contributes: some of the first works to treat error accumulation across timesteps as a problem and proposes a noise-theoretic correction for it. Why it is here: directly informs the understanding of Challenge 2 on quantization error accumulation across the denoising trajectory.*
- [5] Y. Shang et al. Post-Training Quantization on Diffusion Models. *CVPR*, 2023. *What it does: proposes Normally Distributed Timestep Calibration (NDTC), sampling calibration timesteps from a skewed normal distribution to oversample regions of the trajectory where activation distributions are most extreme. What it contributes: some of the early papers identifying timestep-varying activations as a challenge and proposition of a solution. Why it is here: serves as a reference to read and understand its timestep-aware calibration idea that Q-Diffusion, APQ-DM, and our work all build on.*
- [6] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Judd, and S. Han. SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models. *ICML*, 2023. *What it does: proposes a mathematically equivalent per-channel smoothing transformation that migrates quantization difficulty from activations to weights by absorbing a smoothing factor s into the preceding weight matrix. What it contributes: demonstrates that the activation-weight difficulty split can be controlled through a migration hyperparameter α , enabling high accuracy INT8 quantization of large language models. Why it is here: directly related to our per-group SmoothQuant migration idea, which we extend by making α learnable per group to account for the timestep-varying activation-weight difficulty split in diffusion U-Nets.*
- [7] X. Wang et al. Towards Accurate Post-Training Quantization for Diffusion Models. *CVPR*, 2024. *What it does: proposes timestep grouping with group-wise quantization functions for activations and selects optimal calibration timesteps using the structural risk minimization principle. What it contributes: first work to apply a statistical criteria for calibration timestep selection, improving over the heuristic or arbitrary uniform sampling of PTQ4DM and Q-Diffusion for only a single time-step. Why it is here: one of the core works that tests grouping time-steps as an idea so our extends or proposes a different grouping criteria related by the spectral frequency content of x_t rather than activation statistics over time-steps.*
- [8] J. So et al. Temporal Dynamic Quantization for Diffusion Models. *NeurIPS*, 2023. *What it does: trains a lightweight MLP that takes a geometric Fourier encoding of the scalar timestep t as input and outputs a continuous quantization interval, pre-computed into a lookup table at calibration time to introduce zero inference overhead. What it contributes: One of the first proposed methods to replace discrete static scales with a continuous scale function over the denoising trajectory, demonstrating that a learned scale function is also feasible and an option. Why it is here: directly relevant to our per-group MLP scale idea. Our work extends TDQ by conditioning the scale on both t and sample-specific activation statistics $(\mu, \sigma, \max)$ rather than on t alone, so that two samples at the same timestep with different activation magnitudes receive different scales.*

6. Preliminary Implementation Resources

- <https://github.com/hojonathanho/diffusion> Official DDPM implementation by Ho et al. in TensorFlow. Serves as the reference architecture for understanding the U-Net noise estimation network and the forward and reverse diffusion process that all PTQ methods in this work target.

- <https://github.com/Xiuyu-Li/q-diffusion> Official implementation of Q-Diffusion (ICCV 2023). The primary baseline this work builds upon, providing reference code for timestep-aware calibration, split shortcut quantization, and the quantization simulation framework used across CIFAR-10, LSUN, and Stable Diffusion experiments.
- <https://github.com/42Shawn/PTQ4DM> Official implementation of PTQ4DM (CVPR 2023). Useful as a secondary baseline for the NDTC calibration strategy and for reproducing the activation distribution visualizations across timesteps that motivate Challenge 1.
- <https://github.com/ziplab/PTQD> Official implementation of PTQD (NeurIPS 2023); relevant for understanding the quantization noise correction pipeline and mixed-precision per-step bitwidth scheme, which directly informs the error accumulation discussion in Challenge 2 but also as referring baseline possibly.
- <https://github.com/ModelTC/TFMQ-DM> Official implementation of TFMQ-DM (CVPR 2024 Highlight). Serves as another possible baseline.
- <https://github.com/huggingface/diffusers> HuggingFace Diffusers library. Provides pretrained Stable Diffusion models, DDPM and DDIM schedulers, and a U-Net implementation that will be used as the base model for applying and evaluating our proposed PTQ framework.
- <https://github.com/TaylorJocelyn/Diffusion-Model-Quantization> Comprehensive repository collecting all major diffusion model quantization papers and their official code links including PTQ4DM, Q-Diffusion, TFMQ-DM, and others. Used as a reference index for identifying and accessing baseline implementations.
- <https://github.com/mit-han-lab/smoothquant> Official implementation of SmoothQuant (ICML 2023); provides the reference implementation of the per-channel smoothing

transformation and migration hyperparameter α that our per-group learnable SmoothQuant migration extends.